COP2034C Module 1 Programming Assignment

Instructions

For this assignment, you will write a Python program that introduces the Palmer Penguin species. The program will use only basic output statements (the print() function). This assignment will show you how to create a simple program that will give you a quick success while helping you become familiar with the development environment we will use throughout the course.

Development Environment (all of the following steps will be demonstrated in class)

Log in to GitHub and fork the public repository provided by your instructor.

Then complete this assignment in Google Colab (https://colab.research.google.com/).
After your program executes correctly:

1. **Download** your notebook as a Python (.py) file.
2. **Upload** the .py file to your forked GitHub repository.
3. Take a screenshot showing the successful execution of your program in Google Colab and save it as a .png or .jpg file.
4. **Upload** the screenshot to your GitHub repository.

Do not submit the .ipynb notebook. The required submission for this assignment is the Python source file (.py) and the execution screenshot.

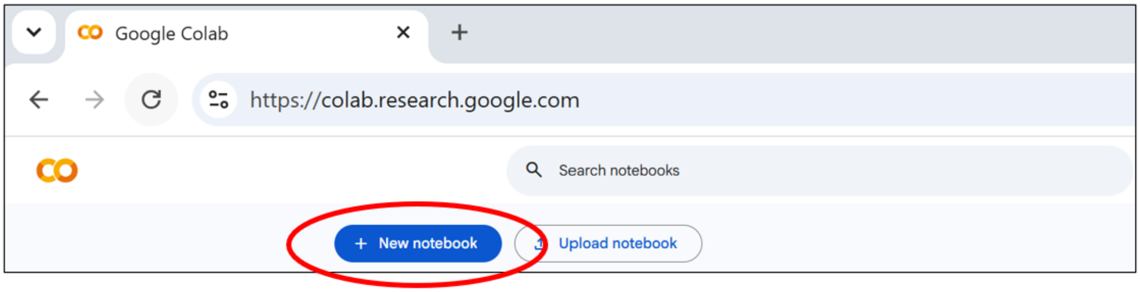
Expected Output

Your program must produce the following output exactly (including capitalization and punctuation).

```
Introducing the Palmer Penguins:
      Chinstrap!
      Gentoo!
and last but not least...
      Adelie!
There are a total of 3 penguin species in this dataset.
```

Walkthrough

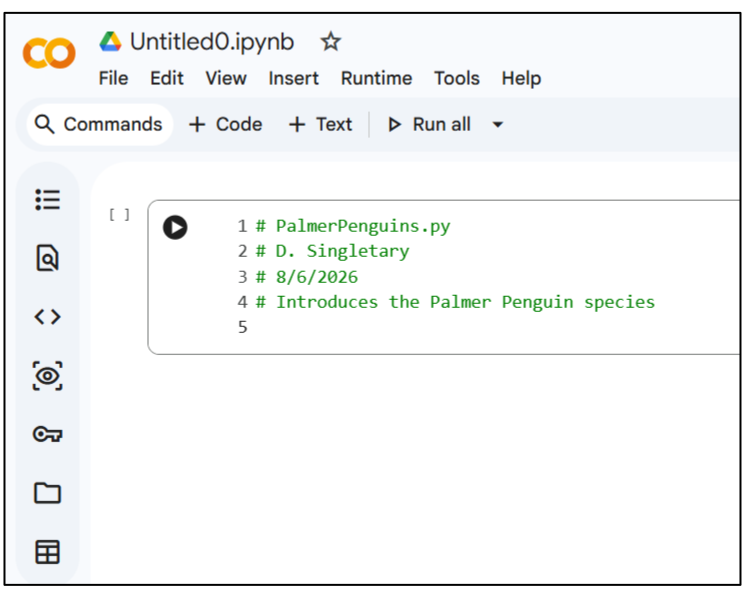
- Create a new notebook in Google Colab.



- Colab opens a new notebook with an empty code cell by default. Any code you enter in this cell can be executed at any time (and as many times as desired, with some restrictions that we will discuss later in the semester). Enter your Python code in this cell. The other type of cell is a text cell, which is used to add formatted notes, explanations, headings, or other documentation to your notebook.
- At the top of the file include the required four-line ID header.

```
# PalmerPenguins.py
# Your Name
# Submission Date
# Introduces the Palmer Penguin species
```

    - Leave a blank line after the header for readability.



Constants

- In Python we commonly represent values that should not change for the life of the program using ALL UPPERCASE variable names. In the following code, we represent the three Palmer Penguin species and the total number of species using Python constants.

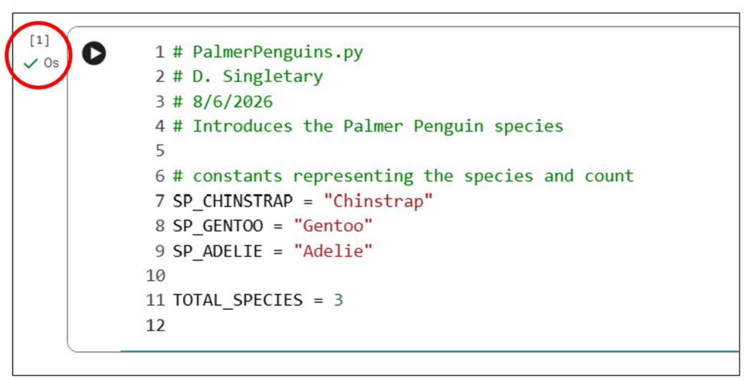
```
# constants representing the species and count
SP_CHINSTRAP = "Chinstrap"
SP_GENTOO = "Gentoo"
SP_ADELIE = "Adelie"

TOTAL_SPECIES = 3
```

- Using these **named constants** makes programs easier to read and maintain.

Incremental Development

- **Develop your program incrementally.** Write a small amount of code, run it, verify that it works, and then continue adding the next piece. Testing frequently makes it much easier to find and correct errors.
- Run the current code cell by clicking the **Run** (▶) button to the left of the cell or by pressing **Shift+Enter**. Although the program does not produce any output yet, running it confirms that your code executes without syntax errors before you continue. In the image below, the red circle highlights the green check mark, execution time, and execution count ([1]), indicating that the cell completed successfully without errors.



Output

- Use only the print() function for this assignment.
- Begin by displaying introductory text.

```
# output the species names with introductory text
print("Introducing the Palmer Penguins:")
```

- Display each penguin species exactly as shown in the expected output.
- You may use the tab escape sequence (\t) for indentation.

```
print("\t" + SP_CHINSTRAP + "!")
```

- Continue with the remaining output.

```
print("\t" + SP_GENTOO + "!")
print("and last but not least...")
print("\t" + SP_ADELIE + "!")
print()
print("There are a total of " +
      str(TOTAL_SPECIES) +
      " penguin species in this dataset.")
```

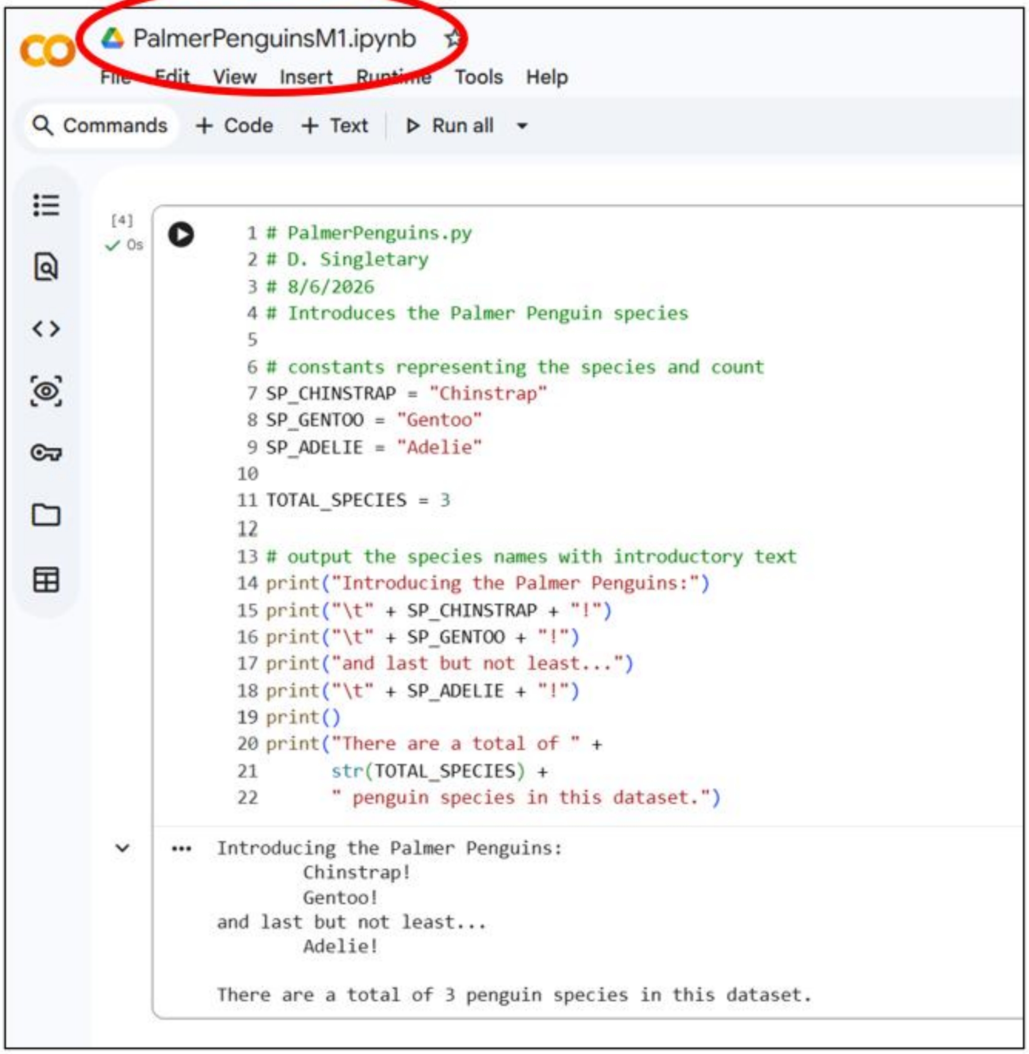
- Notice that TOTAL_SPECIES must be converted to a string using str(TOTAL_SPECIES) before concatenation.
- To print a blank line, we can just use print() with no arguments.

Readability

- In addition to whether your program runs correctly, all assignments in this course will also be graded on code readability. Keep your code readable by:
  - using consistent indentation.
  - separating logical sections with blank lines.
  - keeping lines reasonably short.
  - adding comments where appropriate.
  - choosing meaningful variable names.
  - using whitespace consistently around operators and after commas.

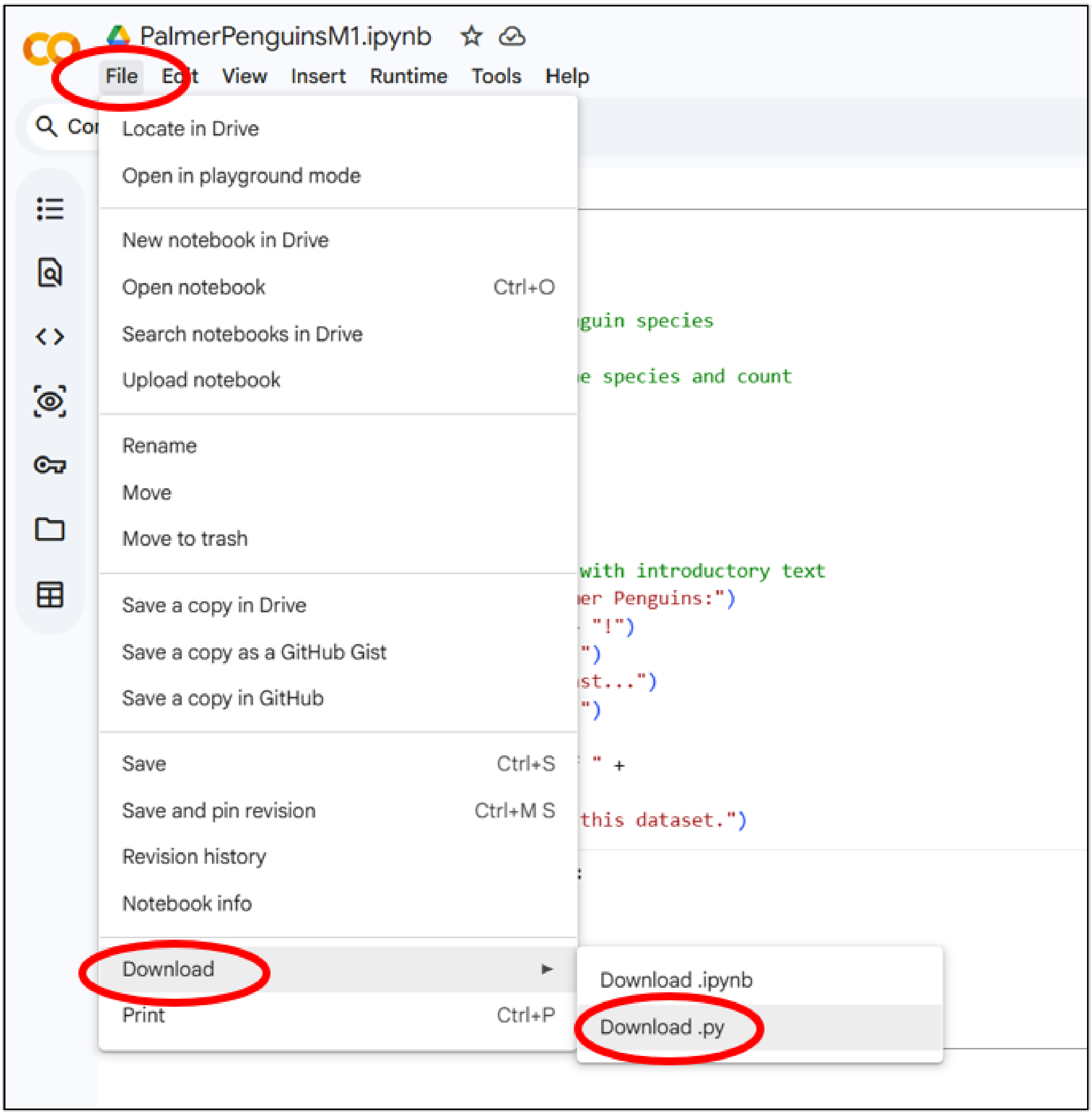
Saving and Downloading

- Give your notebook a name by clicking in the name field and the top and enter the new name:



- Colab notebooks are auto-saved, but you can save them manually by using File/Save in the notebook's menu bar.

- We aren't as interested in saving the notebook as we are in saving the Python code. When your program is complete, download the Python program (do not download or submit the notebook). Download it using the File/Download/Download.py menu option as shown below. Check your downloads folder for the file, be sure it has a meaningful name. Since I called my notebook "PalmerPenguinsM1.ipynb", the downloaded file is called "palmerpenguinsm1.py".



Screenshot

- The final part of this assignment is to show "proof of life", i.e., a screenshot showing your output in the notebook. We are only interested in the output, so here is the screenshot I took with the Windows snipping tool:

```
•••  Introducing the Palmer Penguins:
             Chinstrap!
             Gentoo!
     and last but not least...
             Adelie!

     There are a total of 3 penguin species in this dataset.
```

- Don't include the source code in your image, I will be checking the .py file in your GitHub repo.
- I saved the file as "PalmerPenguinsM1.png".

Submission

Upload the following two files to your forked GitHub repository:

- PalmerPenguins.py
- Screenshot of the program executing successfully (.png or .jpg)

Submit the URL of your forked GitHub repository to the Canvas the assignment.